Department of EEE

MATH 1111 Differential & Integral Calculus

Quiz [20 Points/Marks] Time: 30 Minutes

Answer all of the following questions

Question 1:

(a) Definie Limit of a function at a point. Circuit's transient response is defined by the function $V(t) = \frac{1-e^{-3t}}{sin(5t)}$. Find the limit of V(t) as t approaches 0.

(b) Define continuity of a function at a point. A signal is described by the piecewise function:

$$s(t) = \begin{cases} t^2 + 2c & if \quad t < 1 \\ 5t - c & if \quad t \geq 1 \end{cases}$$

Test the continuity of $S(t)$ at $t = 1$. Find the value of the constant **c** that makes the function s(t) continuous for all real numbers. Rewrite $S(t)$ with the value of c and finally Sketch the graph of S(t).

Question 2:

(a) State Leibnitz's theorem for finding the nth derivative of the product of two functions. Using Leibnitz's theorem, find the nth derivative of the function $y = x^2 e^{ax}$.

(b) If $y = sin(msin^{-1}x)$, use Leibnitz's theorem to prove the following recurrence relation for its derivatives: $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - m^2)y_n = 0$

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Question 1:

(d) Definie Homogeneous Function. Test the homoginity of the following

(i) $f(x,y) = x^2y - 4y^3$ (ii) $g(x,y) = \frac{x}{y^2} + \frac{y}{x^2}$ and (iii) $h(x,y) = x^2 + \sin(y)$

(e) State and Proof the Euler'sTheorem.

Question 2:

(b) Tangent Plane and Normal Line. Find the tangent plane and normal line to the surface $z = 2x^2 + y^2$ at the point (1,1,3)

(f) What is Asymptotes? Define different types of Asymptotes. Find all asymptotes of the curve $y = \frac{x^3}{x^2-1}$.

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Answer any one of the following questions

Question 1:

(a) Gamma function. List fundamental properties of Gamma function.

(b) Prove that for all n > 0, the Gamma function satisfies: $\Gamma(n+1) = n\Gamma(n)$

(c) Define Beta function. By using Beta function evaluate $B(2,3) = \frac{1}{12}$.

Question 2:

(a) Define Multiple Integrals with examples.

(b) Find the area of the region bounded by $y = x^2$ and $y = 2x$.

(c) Find the volume of the solid bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy-plane.

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